

- 5 The curve C has equation $y = \frac{2x^2 - x - 1}{x^2 + x + 1}$.

(a) Show that C has no vertical asymptotes and state the equation of the horizontal asymptote of C . [3]

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(b) Find the coordinates of the stationary points on C . [4]

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- (c) Sketch C , stating the coordinates of the intersections with the axes.

[3]

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- (d) Sketch the curve with equation $y = \left| \frac{2x^2 - x - 1}{x^2 + x + 1} \right|$ and state the set of values of k for which $\left| \frac{2x^2 - x - 1}{x^2 + x + 1} \right| = k$ has 4 distinct real solutions.

[2]

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